



CLEF2022-CheckThat

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Introduction

CLEF2022-CheckThat! task 1a and 1b

Task 1a

- Given a tweet, predict whether it is worth fact-checking. This task is defined with binary labels

Task 1b

- Given a tweet, predict whether it contains a verifiable factual claim. This is a binary task with two labels



Problem statement

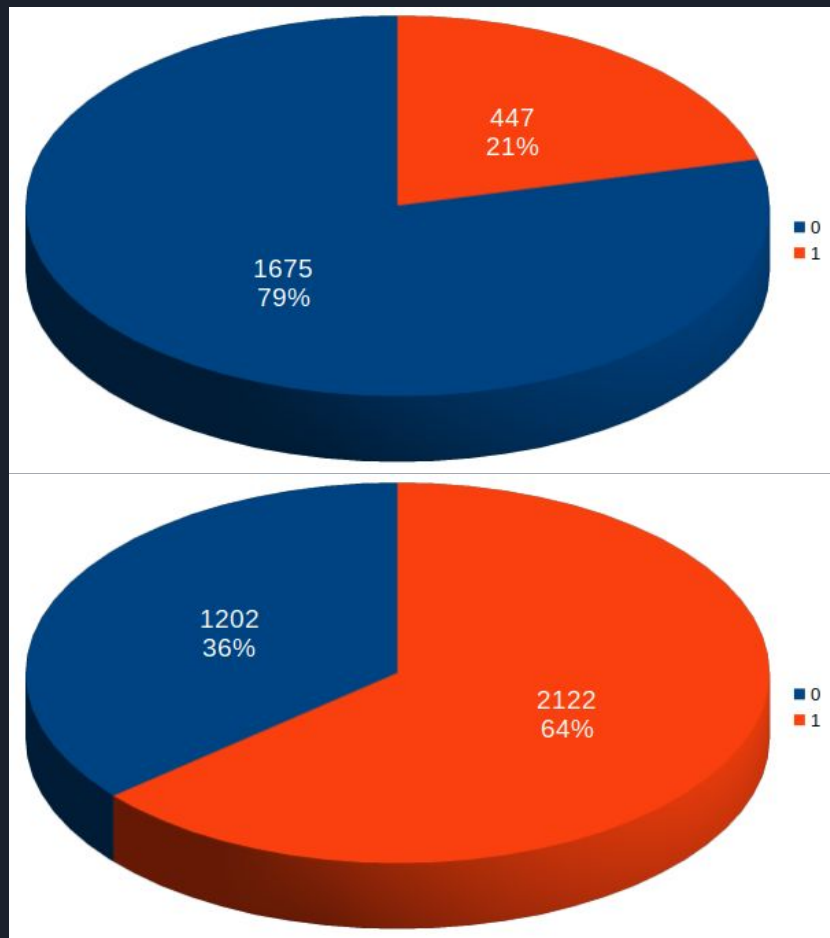
We aim to compare the performance of the LSTM and CNN models in predicting the binary labels to check worthiness and detect verifiable factual claims.

Datasets

The pie chart at the top shows the distribution of the training dataset for task a and at the bottom for task b.

we see clearly the imbalance in the dataset.

- Topic: The topic of the tweet.
- Tweet ID: The unique identifier for the tweet.
- Tweet URL: The URL for the tweet.
- Tweet Text: The text of the tweet.
- Class Label: Label on each tweet whether they are worth fact-checking / if we we could detect verifiable factual claims.





Pre-processing

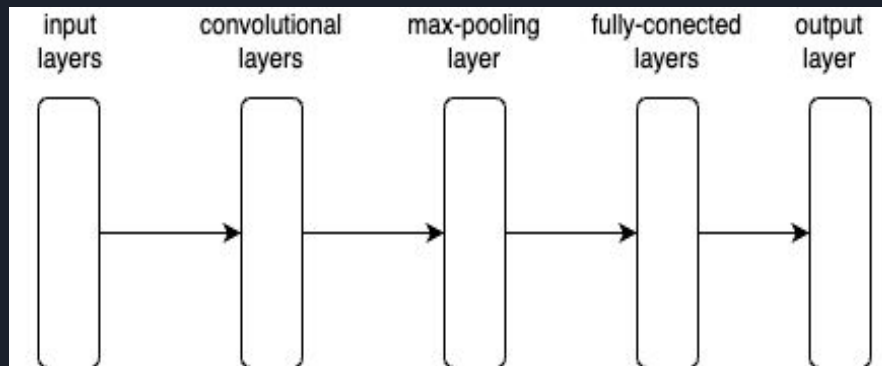
The only features used to train the model is Tweet Text and Class Label. The other features do not have an impact on the results, all the tweets in the dataset have the same topic, the id and url is not important for the task at hand.

To clean the data in the tweet text column we used following pipeline:

- Remove URLs
- Remove mentions and hashtags:
- Remove special characters:
- Remove extra spaces
- Tokenize text and lowering of all characters
- Remove stop words
- Stemming using the Porter stemmer algorithm.

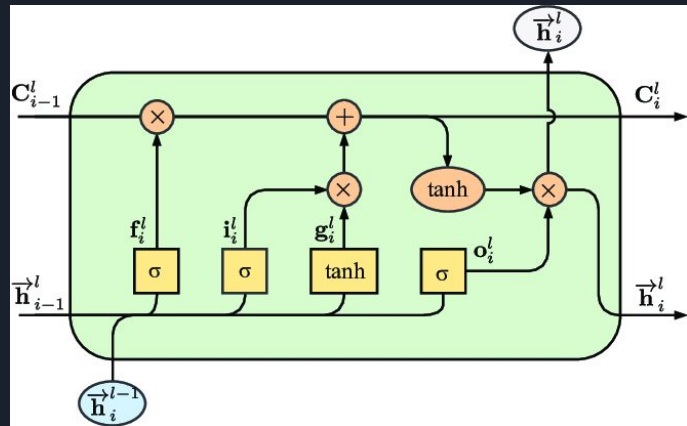
CNN

1. Pretrained embeddings
2. Multiple Convolutional Filters with Different Filter Sizes
3. Non-linear Activation: ReLU
4. Max Pooling
5. Fully Connected Layers
6. Early stopping



LSTM

1. Pretrained Embeddings:
2. Long Short-Term Memory (LSTM) Layer
3. Dropout layer
4. Fully Connected Layer
5. Early stopping



Results after hyperparameter tuning for a

Hidden	Lr	Dropout	Patience	F1
98	0.0005	0.0730	6	0.5183
52	0.000003	0.3247	5	0

Table 1: LMST for checking the worthiness of the tweets

	Max	Min
Filters	130	16
Ngrams	[2]	[6,5,6]
Hidden	29	71
Lr	0.0003	0.000002
Dropout	0.036	0.230
Patience	5	10
F1	0.5364	0

Table 2: CNN for checking the worthiness of the tweets

Results after hyperparameter tuning for b

Hidden	Lr	Dropout	Patience	Accuracy
74	0.0009	0.2079	6	0.7398
75	0.00006	0.4450	8	0.3699

Table 3: LMST for detecting the verifiable factual claims

	Max	Min
Filters	72	64
Ngrams	[7, 3, 5, 2, 2, 3]	[7,3,3]
Hidden	54	53
Lr	0.0006	0.001
Dropout	0.1809	0.2556
Patience	7	7
Accuracy	0.7311	0.6125

Table 4: CNN for detecting the verifiable factual claims



Challenges

- Small dataset
- finding a way to balance the dataset
- Find the best way to create the word embeddings



Future work

- Try different architectures
- Try using a transformer model
- Try ensemble methods

